DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

Project Title: MAX – Sustainable Products E-Commerce Order Management System

1. Aggregate Functions

1.1 COUNT() – Total Number of Orders

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           3
```

1.2 SUM() – Total Sales Amount

```
SQL> SELECT SUM(Subtotal) AS Total_Sales
  2  FROM Order_Details;

TOTAL_SALES
-----------
   11000.00
```

1.3 AVG() – Average Order Amount

```
SQL> SELECT AVG(Order_Amount) AS Average_Order_Amount
  2  FROM (
  3      SELECT Order_ID,
  4             SUM(Subtotal) AS Order_Amount
  5      FROM Order_Details
  6      GROUP BY Order_ID
  7  );

AVERAGE_ORDER_AMOUNT
--------------------
             3666.67
```

1.4 MIN() – Lowest Order Amount

```
SQL> SELECT MIN(Order_Amount) AS Lowest_Order_Amount
  2  FROM (
  3       SELECT Order_ID,
  4              SUM(Subtotal) AS Order_Amount
  5       FROM Order_Details
  6       GROUP BY Order_ID
  7  );

LOWEST_ORDER_AMOUNT
-------------------
            1500.00
```

1.5 MAX() – Highest Order Amount

```
SQL> SELECT MAX(Order_Amount) AS Highest_Order_Amount
  2  FROM (
  3       SELECT Order_ID,
  4              SUM(Subtotal) AS Order_Amount
  5       FROM Order_Details
  6       GROUP BY Order_ID
  7  );

HIGHEST_ORDER_AMOUNT
--------------------
             5000.00
```

2. Generate Total Sales Reports

```
SQL> SELECT SUM(Subtotal) AS Total_Sales
  2  FROM Order_Details;

TOTAL_SALES
-----------
   11000.00
```

2.1 Customer-wise Sales Report

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         SUM(D.Subtotal) AS Total_Purchase
  4  FROM Customer C
  5  JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  JOIN Order_Details D
  8  ON O.Order_ID = D.Order_ID
  9  GROUP BY C.Customer_ID, C.Customer_Name
 10  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME        TOTAL_PURCHASE
----------- -------------------- --------------
          1 Keerthi                    5000.00
          2 Priya                      4500.00
          3 Divya                      1500.00
```

3. Find Top Customers Based on Purchase Amount

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         SUM(D.Subtotal) AS Total_Purchase
  4  FROM Customer C
  5  JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  JOIN Order_Details D
  8  ON O.Order_ID = D.Order_ID
  9  GROUP BY C.Customer_ID, C.Customer_Name
 10  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME        TOTAL_PURCHASE
----------- -------------------- --------------
          1 Keerthi                    5000.00
          2 Priya                      4500.00
          3 Divya                      1500.00
```

4. Identify Best-Selling Products

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         SUM(D.Quantity) AS Total_Quantity_Sold
  4  FROM Product P
  5  JOIN Order_Details D
  6  ON P.Product_ID = D.Product_ID
  7  GROUP BY P.Product_ID, P.Product_Name
  8  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID PRODUCT_NAME                    TOTAL_QUANTITY_SOLD
---------- ------------------------------ -------------------
       105 Sports Shoes                                     3
       101 Women Kurti                                      2
       103 Kids T-Shirt                                     2
       104 Women Jeans                                      1
       102 Men Shirt                                        1
```

4.1 Product Sales Amount

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         SUM(D.Quantity) AS Total_Quantity_Sold,
  4         SUM(D.Subtotal) AS Total_Sales_Amount
  5  FROM Product P
  6  JOIN Order_Details D
  7  ON P.Product_ID = D.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Total_Sales_Amount DESC;

PRODUCT_ID PRODUCT_NAME                    TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
---------- ------------------------------ ------------------- -------------------
       101 Women Kurti                                      2             3000.00
       104 Women Jeans                                      1             2500.00
       102 Men Shirt                                        1             2000.00
       103 Kids T-Shirt                                     2             2000.00
       105 Sports Shoes                                     3             1500.00
```

5. Category-wise Sales Analysis

```
SQL> SELECT P.Category,
  2         SUM(D.Quantity) AS Total_Quantity_Sold,
  3         SUM(D.Subtotal) AS Total_Sales
  4  FROM Product P
  5  JOIN Order_Details D
  6  ON P.Product_ID = D.Product_ID
  7  GROUP BY P.Category
  8  ORDER BY Total_Sales DESC;

CATEGORY             TOTAL_QUANTITY_SOLD TOTAL_SALES
-------------------- ------------------- -----------
Women                                  3     5500.00
Men                                    1     2000.00
Kids                                   2     2000.00
Footwear                               3     1500.00
```

6. Combined Business Sales Report

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         P.Product_ID,
  5         P.Product_Name,
  6         P.Category,
  7         D.Quantity,
  8         D.Unit_Price,
  9         D.Subtotal
 10  FROM Customer C
 11  JOIN Orders O
 12  ON C.Customer_ID = O.Customer_ID
 13  JOIN Order_Details D
 14  ON O.Order_ID = D.Order_ID
 15  JOIN Product P
 16  ON D.Product_ID = P.Product_ID
 17  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME          ORDER_ID PRODUCT_ID PRODUCT_NAME          CATEGORY
       QUANTITY UNIT_PRICE   SUBTOTAL
----------- -------------------- ---------- ---------- ---------------------------- -----
-------------- -------- ---------- ----------
          1 Keerthi                     1001        101 Women Kurti           Women
              2       1500    3000.00
          1 Keerthi                     1001        102 Men Shirt             Men
              1       2000    2000.00
          2 Priya                       1002        103 Kids T-Shirt          Kids
              2       1000    2000.00
          2 Priya                       1002        104 Women Jeans           Women
              1       2500    2500.00
          3 Divya                       1003        105 Sports Shoes          Footwear
              3        500    1500.00
```